



PT. SALIPI NAGARI PERKASA

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LAPORAN UJI MINYAK TRAFO

di

UNIT JASA PEMBANGKIT

UNIT 2

JERANJANG



Yang dilakukan pada tanggal 07 Desember 2017

Untuk keadaan trafo bulan DESEMBER 2017

Untuk pengganti pekerjaan – Mei 2017

TAHAP 2

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1.0 Specific Information:

No.	Unit No.	Serial Number	TC	Test Requested
1.	TRAFO 1 ESP UNIT 2	10037	17	Acidity, Dielectric, Color, Visualization and Water Content
2.	TRAFO 2 ESP UNIT 2	10036	18	Acidity, Dielectric, Color, Visualization and Water Content
3.	TRAFO 3 ESP UNIT 2	10033	19	Acidity, Dielectric, Color, Visualization and Water Content
4.	TRAFO 4 ESP UNIT 2	10032	20	Acidity, Dielectric, Color, Visualization and Water Content
5.	MAIN TRAFO UNIT 2	1006902	21	Acidity, Dielectric, Color, Visualization and Water Content

Note:

* OLTT+ consists of Liquid Screen Tests, Dissolved Gas Analysis (DGA), Water Content, Furan Analysis, Metals-in-Oil Analysis, Liquid Power Factors, and Inhibitor Content (DBPC).

** Liquid Screen Test consists of Acidity, Interfacial Tension (IFT), Dielectric, Colour, Specific Gravity, Visualization, and Sediment.

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2.0 Our Comments and Recommendations of The Test Results Analysis:

We had reviewed each laboratory test result against respective benchmark parameter and following are our comments and recommendations.

No.	Unit No.	Serial Number	TC
1.	TRAFO 1 ESP UNIT 2	10037	17

A) Analysis

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 22kV is **UNACCEPTABLE**. This is reflected from the high moisture content detected in the transformer oil.
- **Colour** of 1.0 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 35ppm is **UNACCEPTABLE** while moisture in solid insulation of 4.13% M/dw is graded as **E**. For this transformer, it is recommended to maintain the water content less than 20ppm. High level of moisture content will act as catalyst to the oxidation process.

B) Recommendation

- Highly recommended to perform **OIL PURIFICATION**. Result indicates that the moisture content in the transformer oil is not conducive for normal operation of a transformer. Oil purification will remove excessive moisture detected in the transformer oil and thus improving dielectric breakdown voltage of the mineral oil. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10037
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	-
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1000
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
28/11/2017	ROUTINE	0.020 AC		48 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.022 AC		22 UN	1.0 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
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ADDITIONAL EQUIPMENT

RADIATORS	-
FANS	-
H2O COOLED	-
OIL PUMPS	-
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	-
CONSERVATOR	-
LTC COMPARTMENT	-
BUSHING LOC	-
FREE BREATHER	-
HOSE LENGTH	-
POWER AVAILABLE	-

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
28/11/2017	27	21	2.48	D
06/12/2017	27	35	4.13	E

FIELD SERVICE

DATE	SERVICE
------	---------

RECOMMENDATION:

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG - FUJIAN
DATE MFG - 2010
MAIN TANK S/N - **10037**
KVA -
PRI - 72000 V
SEC - 380 V
KILOGRAMS -
EQUIP TYPE - TRANSFORMER
TRANS CLASS - ONAN
IMPEDENCE -
PHASE/CYCLE - 3/50 Hz
LIQUID TYPE - OIL
LITRES LIQUID - 1000

LTC MFG -
LTC S/N -
LTC FAB NO -
LTC MODEL -
LTC LITRES LIQUID -
LTC ENERGIZED -

ADDITIONAL EQUIPMENT

RADIATORS -
FANS -
H2O COOLED -
OIL PUMPS -
TOP FPV -
BOTTOM FPV -
BUCHOLZ -
CONSERVATOR -
LTC COMPARTMENT -
BUSHING LOC -
FREE BREATHER -
HOSE LENGTH -
POWER AVAILABLE -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE 5H2F 2FOL 2FAL 2ACF 5M2F TOTAL

RECOMMENDATION:

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE [®]ALUMINIUM [®]IRON COPPER

LIQUID POWER FACTOR (ASTM D924)

DATE 25°C 100°C

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE %BY WEIGHT

[®]CORROSIVE SULFUR (ASTM 1275B)

DATE STATUS CLASS APPEARANCE

[®]PASSIVATOR COUNT, EXPRESSED IN PPM

DATE PPM

FIELD SERVICE

DATE SERVICE

No.	Unit No.	Serial Number	TC
2.	TRAFO 2 ESP UNIT 2	10036	18

A) Analysis

- **Acidity** of 0.051mgKOH/gm is **QUESTIONABLE**. This indicates the starting formation of oxidation by products in the oil.
- **Dielectric** of 59kV is **ACCEPTABLE**.
- **Colour** of 1.0 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 14ppm is **ACCEPTABLE** while moisture in solid insulation of 1.65% M/dw is graded as **C**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Retest **Acidity** and **Interfacial Tension** in 6 months to assess if the insulating oil deteriorate to a level that require corrective action to improve the oil quality.
- Other parameters: Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10036
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	-
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1000
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
28/11/2017	ROUTINE	0.048 AC		55 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.051 QU		59 AC	1.0 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
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ADDITIONAL EQUIPMENT

RADIATORS	-
FANS	-
H2O COOLED	-
OIL PUMPS	-
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	-
CONSERVATOR	-
LTC COMPARTMENT	-
BUSHING LOC	-
FREE BREATHER	-
HOSE LENGTH	-
POWER AVAILABLE	-

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
28/11/2017	28	13	1.53	C
06/12/2017	28	14	1.65	C

FIELD SERVICE

DATE	SERVICE
------	---------

RECOMMENDATION:

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10036
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	-
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1000
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
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RECOMMENDATION:

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
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LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
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INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
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ADDITIONAL EQUIPMENT

RADIATORS	-
FANS	-
H2O COOLED	-
OIL PUMPS	-
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	-
CONSERVATOR	-
LTC COMPARTMENT	-
BUSHING LOC	-
FREE BREATHER	-
HOSE LENGTH	-
POWER AVAILABLE	-

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
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®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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No.	Unit No.	Serial Number	TC
3.	TRAFO 3 ESP UNIT 2	10033	19

A) Analysis

- **Acidity** of 0.056mgKOH/g is **QUESTIONABLE**. This indicates the presence of oxidation by products in the oil. Close monitoring is needed.
- **Dielectric** of 60kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 16ppm is **ACCEPTABLE** while moisture in solid insulation of 1.89% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Retest **Acidity** and **Interfacial Tension** in 6 months to assess if the insulating oil deteriorate to a level that require corrective action to improve the oil quality.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10033
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	-
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1000
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
28/11/2017	ROUTINE	0.053 QU		42 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.056 QU		60 AC	1.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
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ADDITIONAL EQUIPMENT

RADIATORS	-
FANS	-
H2O COOLED	-
OIL PUMPS	-
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	-
CONSERVATOR	-
LTC COMPARTMENT	-
BUSHING LOC	-
FREE BREATHER	-
HOSE LENGTH	-
POWER AVAILABLE	-

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
28/11/2017	28	15	1.77	D
06/12/2017	28	16	1.89	D

FIELD SERVICE

DATE	SERVICE
------	---------

RECOMMENDATION:

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG - FUJIAN
DATE MFG - 2010
MAIN TANK S/N - **10033**
KVA -
PRI - 72000 V
SEC - 380 V
KILOGRAMS -
EQUIP TYPE - TRANSFORMER
TRANS CLASS - ONAN
IMPEDENCE -
PHASE/CYCLE - 3/50 Hz
LIQUID TYPE - OIL
LITRES LIQUID - 1000

LTC MFG -
LTC S/N -
LTC FAB NO -
LTC MODEL -
LTC LITRES LIQUID -
LTC ENERGIZED -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE 5H2F 2FOL 2FAL 2ACF 5M2F TOTAL

RECOMMENDATION:

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE [®]ALUMINIUM [®]IRON COPPER

LIQUID POWER FACTOR (ASTM D924)

DATE 25°C 100°C

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE %BY WEIGHT

ADDITIONAL EQUIPMENT

RADIATORS -
FANS -
H2O COOLED -
OIL PUMPS -
TOP FPV -
BOTTOM FPV -
BUCHOLZ -
CONSERVATOR -
LTC COMPARTMENT -
BUSHING LOC -
FREE BREATHER -
HOSE LENGTH -
POWER AVAILABLE -

[®]CORROSIVE SULFUR (ASTM 1275B)

DATE STATUS CLASS APPEARANCE

[®]PASSIVATOR COUNT, EXPRESSED IN PPM

DATE PPM

FIELD SERVICE

DATE SERVICE

No.	Unit No.	Serial Number	TC
4.	TRAFO 4 ESP UNIT 2	10032	20

A) Analysis

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 58kV is **ACCEPTABLE**.
- **Colour** of 1.0 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 17ppm is **ACCEPTABLE** while moisture in solid insulation of 2.01% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10032
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	-
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1000
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
28/11/2017	ROUTINE	0.020 AC		62 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.022 AC		58 AC	1.0 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
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ADDITIONAL EQUIPMENT

RADIATORS	-
FANS	-
H2O COOLED	-
OIL PUMPS	-
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	-
CONSERVATOR	-
LTC COMPARTMENT	-
BUSHING LOC	-
FREE BREATHER	-
HOSE LENGTH	-
POWER AVAILABLE	-

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
28/11/2017	28	16	1.89	D
06/12/2017	28	17	2.01	D

FIELD SERVICE

DATE	SERVICE
------	---------

RECOMMENDATION:

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10032
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	-
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1000
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
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RECOMMENDATION:

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
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LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
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INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
------	------------

ADDITIONAL EQUIPMENT

RADIATORS	-
FANS	-
H2O COOLED	-
OIL PUMPS	-
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	-
CONSERVATOR	-
LTC COMPARTMENT	-
BUSHING LOC	-
FREE BREATHER	-
HOSE LENGTH	-
POWER AVAILABLE	-

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
------	--------	-------	------------

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
5.	MAIN TRAF0 UNIT 2	1006902	21

A) Analysis

- **Acidity** of 0.008mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 67kV is **ACCEPTABLE**.
- **Colour** of 0.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 13ppm is **ACCEPTABLE** while moisture in solid insulation of 0.85% M/dw is graded as **A**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAFU UNIT 2

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
27/11/2017	ONLINE	50	55				
06/12/2017	ONLINE						

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
27/11/2017	ROUTINE	0.008 AC		70 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.008 AC		67 AC	0.5 AC		CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
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RECOMMENDATION:

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
27/11/2017	37	16	1.04	B
06/12/2017	38	13	0.85	A

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 1006902
KVA	- 33000
PRI	- 170000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.03%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 11160
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	-
FANS	-
H2O COOLED	-
OIL PUMPS	-
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	-
CONSERVATOR	-
LTC COMPARTMENT	-
BUSHING LOC	-
FREE BREATHER	-
HOSE LENGTH	-
POWER AVAILABLE	-

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAFU UNIT 2

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 1006902
KVA	- 33000
PRI	- 170000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.03%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 11160
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	-
FANS	-
H2O COOLED	-
OIL PUMPS	-
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	-
CONSERVATOR	-
LTC COMPARTMENT	-
BUSHING LOC	-
FREE BREATHER	-
HOSE LENGTH	-
POWER AVAILABLE	-

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
06/12/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
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LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
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INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
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CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
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PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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3.0 Summary of Our Comments & Recommendations:

No.	TX Unit No.	Serial No.	TC	Recommendation
1.	TRAFO 1 ESP UNIT 2	10037	17	• Perform OIL PURIFICATION
2.	TRAFO 2 ESP UNIT 2	10036	18	• Retest 6 months: Acidity and IFT • Other parameters: Periodic condition monitoring
3.	TRAFO 3 ESP UNIT 2	10033	19	• Retest 6 months: Acidity and IFT • Other parameters: Periodic condition monitoring
4.	TRAFO 4 ESP UNIT 2	10032	20	• Periodic condition monitoring
5.	MAIN TRAFO UNIT 2	1006902	21	• Periodic condition monitoring

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein



no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test laboratorium cennix								MALAYSIAN LAB RECOMMENDATION		
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture			
															ppm			grade
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533			
		BATASAN OPERASI						< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20			
1	17	27/11/17	Trafo 1 ESP	2	10037	1010	731	Fujian	0,020		48	1,0		C		21	D	Retest 6 months: Dielectric and Water Content
	17	07/12/17							0,022		22	1,0		C		35	E	OIL PURIFICATION
2	18	27/11/17	Trafo 2 ESP	2	10036	1010	731	Fujian	0,048		55	1,0		C		13	C	Periodic condition monitoring
	18	07/12/17							0,051		59	1,0		C		14	C	Retest 6 months: Acidity and IFT Periodic condition monitoring
3	19	27/11/17	Trafo 3 ESP	2	10033	1010	731	Fujian	0,053		42	1,5		C		15	D	Retest 6 months: Acidity and IFT Other parameters: Periodic Condition Monitoring
	19	07/12/17							0,056		60	1,5		C		16	D	Retest 6 months: Acidity and IFT Other parameters: Periodic Condition Monitoring
4	20	27/11/17	Trafo 4 ESP	2	10032	1010	731	Fujian	0,020		62	1,0		C		16	D	Periodic condition monitoring
	20	07/12/17							0,022		58	1,0		C		17	D	
5	21	27/11/17	Main Trafo	2	1006902	2011	11.160	Wolong	0,008		70	0,5		C		16	B	Periodic condition monitoring
	21	07/12/17							0,008		67	0,5		C		13	A	Periodic condition monitoring

Summary by Soetjahjono
TxM Indonesia

PAK SOETJAHJONO (TXM ADVISOR)
RECOMMENDATION

Trafo ESP 1 mengalami percepatan penurunan Grade Moisture yang artinya kondisi air dalam isolasi kertas semakin tinggi. Disarankan untuk segera dilakukan Purifikasi dengan temperatur tinggi agar kandungan air dalam isolasi kertas dapat ikut keluar.

Kondisi semua **trafo ESP (Tc 17, 18, 19, 20)** Grade Moisture rendah, yang artinya kandungan air dalam isolasi kertas cukup tinggi. Kami sarankan untuk dimonitor dengan test DGA agar bisa diantisipasi jika ada kondisi gangguan thermal yang terjadi pada trafo trafo tersebut.